

From Alternating Bounds to Rational Error Estimates: A Padé–Continued Fraction Framework

Zihan (Aulinda) Wei

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Abstract

Alternating series, continued fractions, and Padé approximations are often introduced as separate methods of approximation. This paper asserts that they are more naturally understood as three levels of the same idea. Alternating series give the simplest form of bracketing, where the error is controlled by the next term. Continued fractions reorganize the same alternating behavior into rational approximations, where the error is controlled by denominator growth. Padé approximations then provide the general rational framework in which ordinary partial sums and continued fraction convergents appear as special cases. After comparing these methods through the example of $\ln 2$, this paper proposes a hybrid Padé-bracket error estimate: estimate it by the gap between adjacent rational approximations. The proposed estimate is rigorous when the approximations form a true bracket and useful as a numerical estimator when such a bracket is not known in advance.

1 Introduction

One of the most familiar ways to approximate a number is to truncate an infinite series. For example,

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots.$$

The alternating series estimate tells us that if we stop after the n th term, then the error is no larger than the next unused term. This is a beautiful result because it is simple, stable, and easy to apply. It also has a clear geometric meaning: the partial sums approach the true value from alternating sides.

However, this simplicity comes with a cost. The classical alternating estimate uses only local information. It sees the next term, but it does not see the larger analytic structure of the function being approximated. For $\ln 2$, the estimate

$$|R_n| \leq \frac{1}{n+1}$$

means that reaching an error below 10^{-4} requires roughly ten thousand terms. The problem is not that $\ln 2$ is impossible to approximate. The problem is that our regular alternating series error bound is an insufficient approximation method.

This paper studies how that insufficiency can be improved by moving from polynomial approximation to rational approximation. The main thesis is that the alternating series estimate is not a phenomenon separate from continued fractions or Padé approximations. Instead, it is the simplest case of a broader rational bracketing principle. Alternating series control error through the next

term; continued fractions control error through denominator growth; Padé approximations explain both methods as special paths inside a general table of rational approximations.

The final goal is to propose a hybrid estimate, called the Padé-bracket estimate. In its simplest form, it replaces the classical next-term bound

$$|S - S_n| \leq |S_{n+1} - S_n|$$

with a rational version:

$$|f(x) - R_N(x)| \lesssim |R_N(x) - R_{N-1}(x)|,$$

where $R_N(x)$ and $R_{N-1}(x)$ are adjacent Padé-type rational approximations. The estimate will become a rigorous inequality when these rational approximations bracket the true value. Even when bracketing is not known, it will still give a practical numerical error indicator.

2 Alternating Series as Linear Bracketing

We begin with the classical alternating series test. Let

$$S = \sum_{k=0}^{\infty} (-1)^k a_k,$$

where

$$a_k > 0, \quad a_{k+1} \leq a_k, \quad \lim_{k \rightarrow \infty} a_k = 0.$$

Define the n th partial sum by

$$S_n = \sum_{k=0}^n (-1)^k a_k.$$

Then the partial sums converge to S , and the error satisfies

$$|S - S_n| \leq a_{n+1}.$$

More importantly, the partial sums do not simply get close to S ; they bracket it. With the sign convention above, the even partial sums approach from one side and the odd partial sums approach from the other:

$$S_1 \leq S_3 \leq S_5 \leq \cdots \leq S \leq \cdots \leq S_4 \leq S_2 \leq S_0.$$

Thus alternating series approximation is not only an approximation method. It is also a bracketing method.

The error estimate follows directly from the bracketing structure. Since S lies between S_n and S_{n+1} , we have

$$|S - S_n| \leq |S_{n+1} - S_n| = a_{n+1}.$$

This interpretation is important because it shows that the usual next-term estimate is really an adjacent-approximations estimate. The next term matters because it is exactly the distance between two consecutive partial sums.

But ordinary alternating sums are still limited. The partial sum

$$S_n(x) = \sum_{k=0}^n (-1)^k a_k x^k$$

is a polynomial approximation. It keeps adding terms, but it does not change the previous terms. In rational approximation language, it is equivalent to using a denominator equal to 1:

$$S_n(x) = \frac{P_n(x)}{1}.$$

This observation will become important later, because it means that ordinary series truncation is a degenerate rational approximation.

3 Continued Fractions as Rational Bracketing

Continued fractions improve the situation by replacing polynomial truncation with rational truncation. A simple continued fraction has the form

$$x = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \cdots}}},$$

where the a_i are positive integers for $i \geq 1$. Its n th convergent is written

$$\frac{p_n}{q_n}.$$

The numerators and denominators satisfy the recurrences

$$p_n = a_n p_{n-1} + p_{n-2}, \quad q_n = a_n q_{n-1} + q_{n-2},$$

with the usual initial conditions

$$p_{-2} = 0, \quad p_{-1} = 1, \quad q_{-2} = 1, \quad q_{-1} = 0.$$

The key difference is that a continued fraction does not solely attach a new term to the end of a sum. Each new convergent updates both the numerator and denominator. The approximation has the form

$$\frac{p_n}{q_n},$$

so the denominator becomes part of the approximation mechanism.

For simple continued fractions, adjacent convergents satisfy the determinant identity

$$p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}.$$

Dividing by $q_n q_{n-1}$ gives

$$\frac{p_n}{q_n} - \frac{p_{n-1}}{q_{n-1}} = \frac{(-1)^{n-1}}{q_n q_{n-1}}.$$

This identity says that the gaps between consecutive convergents alternate in sign, and the size of the gap is controlled by a product of denominators.

In particular, the convergents themselves can be written as a kind of alternating sum:

$$\frac{p_n}{q_n} = \frac{p_0}{q_0} + \sum_{k=1}^n \left(\frac{p_k}{q_k} - \frac{p_{k-1}}{q_{k-1}} \right) = \frac{p_0}{q_0} + \sum_{k=1}^n \frac{(-1)^{k-1}}{q_k q_{k-1}}.$$

This does not mean that every continued fraction comes from an ordinary alternating series. Rather, it shows that the sequence of convergents has its own alternating structure: consecutive rational approximations move from opposite sides, and their gaps are controlled by denominator products.

The standard error bound for continued fractions also reflects this structure:

$$\left| x - \frac{p_n}{q_n} \right| < \frac{1}{q_n q_{n+1}}.$$

This should be compared with the alternating series error bound:

$$|S - S_n| \leq a_{n+1}.$$

Thus, the continued fraction estimate plays the same role as the alternating-series estimate: it explains why the current approximation is close to the true value. The difference is that the alternating-series estimate measures this closeness by the next unused term, while the continued-fraction estimate measures it by the growth of the rational denominators.

4 Padé approximations as the General Framework

Padé approximations allow us to unify the previous two methods. Let

$$f(x) = c_0 + c_1 x + c_2 x^2 + \cdots$$

be a formal power series. The ordinary Taylor polynomial of degree n is

$$T_n(x) = c_0 + c_1 x + \cdots + c_n x^n.$$

A Padé approximation replaces this polynomial by a rational function.

Definition 1. *The Padé approximations of type $[L/M]$ to f is a rational function*

$$\frac{P_L(x)}{Q_M(x)},$$

where

$$\deg P_L \leq L, \quad \deg Q_M \leq M,$$

usually normalized by $Q_M(0) = 1$, such that

$$f(x) - \frac{P_L(x)}{Q_M(x)} = O(x^{L+M+1}).$$

Equivalently,

$$Q_M(x)f(x) - P_L(x) = O(x^{L+M+1}).$$

This definition shows why Padé approximation is stronger than ordinary truncation. A Taylor polynomial chooses a numerator and fixes the denominator as 1. A Padé approximation chooses both numerator and denominator so that as many low-order terms as possible cancel.

Note that ordinary partial sums are also included in this framework. If $M = 0$, then $Q_0(x) = 1$, and the Padé approximation $[n/0]$ is just the Taylor polynomial of degree n :

$$[n/0] = T_n(x).$$

Continued fractions are also connected to Padé approximations. Many continued fractions generated from power series have convergents that move through the Padé table, such as

$$[0/0], \quad [1/0], \quad [1/1], \quad [2/1], \quad [2/2], \quad [3/2], \quad [3/3], \dots$$

This means that continued fraction convergents are not alone another approximation method. They can be understood as a sequence of rational approximations inside the Padé table.

The relationship can be summarized as follows:

$$\begin{aligned} &\boxed{\text{ordinary partial sums} = [n/0] \text{ Padé approximations}}, \\ &\boxed{\text{continued fraction convergents} = \text{staircase Padé approximations}}, \\ &\boxed{\text{diagonal Padé approximations } [n/n] = \text{balanced rational approximations}}. \end{aligned}$$

Therefore alternating series, continued fractions, and Padé approximations should not be seen as separate tools.

5 Three Ways to Interpret Error

The three approximation methods differ mainly in how they interpret error.

For an alternating series,

$$R_n = S - S_n = (-1)^{n+1}a_{n+1} + (-1)^{n+2}a_{n+2} + \dots$$

The error is just the remaining tail of the series, so the natural estimate is

$$|R_n| \leq a_{n+1}.$$

This is a tail estimate.

For a continued fraction, the error is not measured by the next term of the original series. Instead, the convergents form rational brackets, and their gaps are controlled by denominator products:

$$\left| x - \frac{p_n}{q_n} \right| < \frac{1}{q_n q_{n+1}}.$$

This is a denominator-growth estimate.

For a Padé approximations, the defining condition is

$$Q_M(x)f(x) - P_L(x) = O(x^{L+M+1}).$$

Thus

$$f(x) - \frac{P_L(x)}{Q_M(x)} = \frac{Q_M(x)f(x) - P_L(x)}{Q_M(x)}.$$

In other words,

$$\text{Padé error} = \frac{\text{high-order residual}}{\text{denominator correction}}.$$

The denominator in a rational approximation is not just a formal addition. By multiplying $f(x)$ in the residual $Q_M(x)f(x) - P_L(x)$, it helps cancel low-order error terms and reorganize the remaining error into a rational form.

This is the main conceptual reason Padé approximations often converge faster than ordinary series truncations. A Taylor polynomial can only match the series by adding more powers of x . A Padé approximations can use a denominator to imitate behavior that a polynomial would need many more terms to.

6 Case Study: Approximating $\ln 2$

We now compare the three viewpoints using

$$\ln 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots.$$

6.1 The alternating harmonic series

The alternating estimate gives

$$|R_n| \leq \frac{1}{n+1}.$$

To guarantee error below 10^{-4} , we need

$$\frac{1}{n+1} < 10^{-4},$$

so

$$n > 9999.$$

This is mathematically reliable and indeed practiced bracketing structure, but computationally inefficient.

6.2 Padé approximations for $\ln(1+x)$

Now consider the power series

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots.$$

The ordinary second-degree Taylor approximation at $x = 1$ is

$$1 - \frac{1}{2} = \frac{1}{2},$$

which is far from

$$\ln 2 \approx 0.693147.$$

Now construct the Padé approximant $[1/1]$. Write

$$[1/1](x) = \frac{ax}{1+bx}.$$

Expanding gives

$$\frac{ax}{1+bx} = ax - abx^2 + ab^2x^3 - \cdots.$$

Matching the first two nonzero terms of $\ln(1+x)$ gives

$$a = 1, \quad -ab = -\frac{1}{2},$$

so

$$b = \frac{1}{2}.$$

Therefore

$$[1/1](x) = \frac{x}{1+x/2} = \frac{2x}{x+2}.$$

At $x = 1$, this gives

$$[1/1](1) = \frac{2}{3} \approx 0.666667.$$

This is already much better than the ordinary truncation $1 - \frac{1}{2} = 0.5$, even though both approximations use only low-order information from the same series.

Higher diagonal Padé approximations improve quickly. For example,

$$[2/2](x) = \frac{x(x+2)/2}{(x^2+6x+6)/6},$$

so

$$[2/2](1) = \frac{9}{13} \approx 0.692308.$$

Also,

$$[3/3](x) = \frac{x(11x^2+60x+60)/60}{(x+2)(x^2+10x+10)/20},$$

so

$$[3/3](1) \approx 0.693122.$$

The comparison is shown below.

Method	Approximation	Value	Absolute error
Taylor $T_1(1)$	1	1.000000	0.306853
Taylor $T_2(1)$	$1 - \frac{1}{2}$	0.500000	0.193147
Padé $[1/1](1)$	$\frac{2}{3}$	0.666667	0.026481
Padé $[2/2](1)$	$\frac{9}{13}$	0.692308	0.000839
Padé $[3/3](1)$	$\frac{131}{189}$	0.693122	0.000025

This table shows the main point of the paper. The Padé approximations do not use a different function. They use the same local series information, but they organize it as a rational function. The denominator lets the approximation respond to the shape of $\ln(1+x)$ more efficiently than a polynomial truncation can.

7 A Proposed Hybrid Estimate: The Padé-Bracket Error

The previous sections suggest a natural hybrid method. In an alternating series, the error is bounded by the gap between adjacent partial sums:

$$|S - S_n| \leq |S_{n+1} - S_n|.$$

It indeed works since consecutive partial sums bracket the true value.

Continued fractions use the same idea in rational form. Consecutive convergents lie on alternating sides of the target, and

$$\left| \frac{p_n}{q_n} - \frac{p_{n-1}}{q_{n-1}} \right| = \frac{1}{q_n q_{n-1}}.$$

Thus the gap between adjacent rational approximations gives an error scale.

This motivates the following definition.

Definition 2 (Padé-bracket error estimate). *Let $R_N(x)$ and $R_{N-1}(x)$ be adjacent rational approximations along a chosen Padé path, for example a diagonal path, a near-diagonal path, or a staircase path coming from a continued fraction. Define*

$$E_N^{PB}(x) = |R_N(x) - R_{N-1}(x)|.$$

If

$$R_{N-1}(x) \leq f(x) \leq R_N(x)$$

or

$$R_N(x) \leq f(x) \leq R_{N-1}(x),$$

then

$$|f(x) - R_N(x)| \leq E_N^{PB}(x).$$

In a more explicit rational form, if

$$R_N(x) = \frac{P_N(x)}{Q_N(x)}, \quad R_{N-1}(x) = \frac{P_{N-1}(x)}{Q_{N-1}(x)},$$

then

$$E_N^{PB}(x) = \left| \frac{P_N(x)}{Q_N(x)} - \frac{P_{N-1}(x)}{Q_{N-1}(x)} \right|.$$

This is the rational analogue of the alternating series next-term estimate. The ordinary alternating estimate is recovered when

$$R_N = S_N, \quad R_{N-1} = S_{N-1}.$$

The continued fraction estimate is recovered when

$$R_N = \frac{p_N}{q_N}, \quad R_{N-1} = \frac{p_{N-1}}{q_{N-1}},$$

because then

$$E_N^{PB} = \left| \frac{p_N}{q_N} - \frac{p_{N-1}}{q_{N-1}} \right| = \frac{1}{q_N q_{N-1}}.$$

Therefore the Padé-bracket estimate merges the three main ideas of this paper:

alternating bracketing $\longrightarrow$ continued-fraction denominator growth $\longrightarrow$ Padé adjacent-rational error estimate.

There is one important limitation. The inequality

$$|f(x) - R_N(x)| \leq E_N^{PB}(x)$$

is only guaranteed when the adjacent rational approximations actually bracket $f(x)$. For some classes of functions, such as many Stieltjes functions, Padé approximations have known bracketing behavior. For a general function, however, this bracketing must be proved or numerically checked. In that case, $E_N^{PB}(x)$ should be described as an estimator rather than a theorem.

This limitation is not a weakness of the method. The classical alternating estimate also depends on a bracketing condition: decreasing positive terms with alternating signs. The Padé-bracket estimate is the same philosophy lifted into rational approximation.

8 Conclusion

Alternating series, continued fractions, and Padé approximations are often taught as separate methods, but they fit into a single approximation story. Alternating series give the first and simplest error model: the true value lies between adjacent partial sums, so the error is bounded by the next term. Continued fractions keep the alternating bracketing behavior but reorganize it into rational approximations, where the error is controlled by products of denominators. Padé approximations then give the general algebraic framework: ordinary partial sums are edge cases $[n/0]$, continued fraction convergents often follow staircase paths, and diagonal approximations give balanced rational approximations.

The proposed Padé-bracket estimate follows from this unified view. Instead of estimating error by the next unused term of a series, it estimates error by the gap between adjacent rational approximations:

$$E_N^{PB}(x) = |R_N(x) - R_{N-1}(x)|.$$

When those approximations bracket the true value, this gives a rigorous error bound. When they do not have a proved bracketing property, it still gives a useful numerical estimate. In this sense, the Padé-bracket estimate is not a complete invention, but a natural extension of the alternating series idea into the rational world.